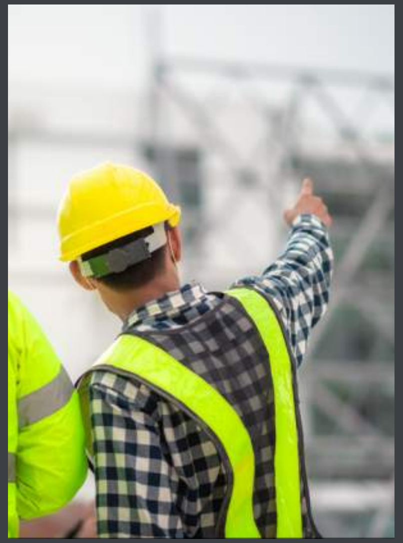
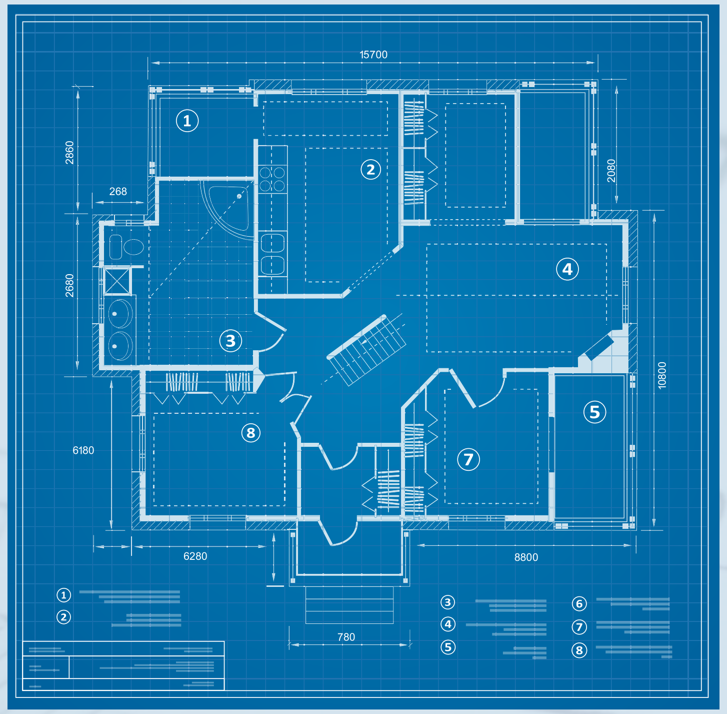
CONSTRUCTION QUALITY MANAGEMENT





- What is a QA system
- Project QA vs Construction QA
- Link between design and quality
- Commissioning
- Components
- The biggest mistake

CONSTRUCTION QA

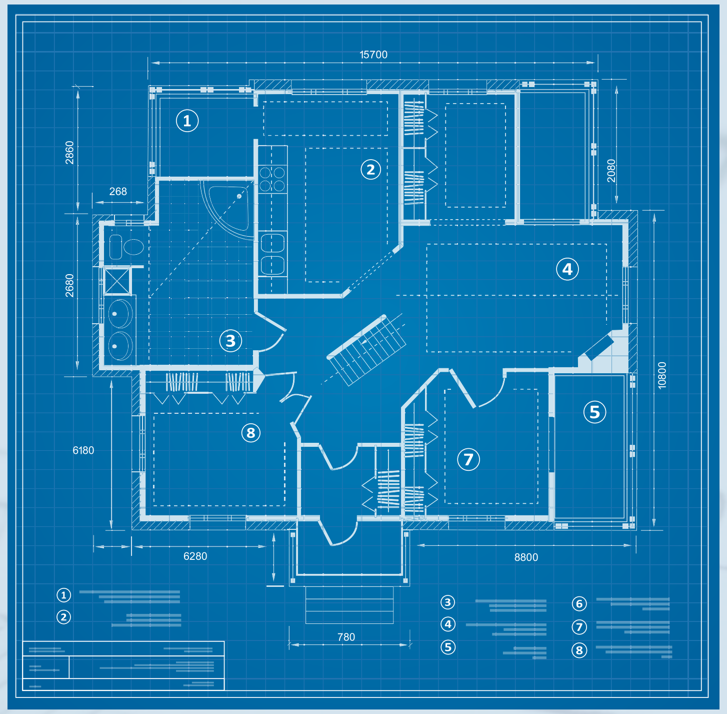


Drawings

Construction

Finished Product

CONSTRUCTION QA



Drawings

Construction

Finished Product

QA System - Ensures the works comply with the drawings and specifications

PROJECT QUALITY VS CONSTRUCTION QUALITY

- Project quality - ensure project fulfils requirements
- Construction quality - ensures what we build complies with drawings and specifications
- The difference?
  - Construction quality assumes design captures project requirements
  - Comply with design, comply with project requirements

DESIGN DEVELOPMENT

- Design transfers stakeholder/client requirements into drawings and specifications
    - “We want a 10 story building”
    - “The building looks like this”
    - “The building code requires fire extinguishers”

CONSTRUCTION QUALITY

- Design develops technical solution to fulfill stakeholder requirements
- Therefore, if we build what the design documents, we fulfill stakeholder requiremnets

DESIGN QUALITY

- How do we ensure the design is correct?
- Three core processes
  - Quality assurance during the design phase
  - Identification of errors and issues during construction
  - Commissioning

QA DURING DESIGN DEVELOPMENT

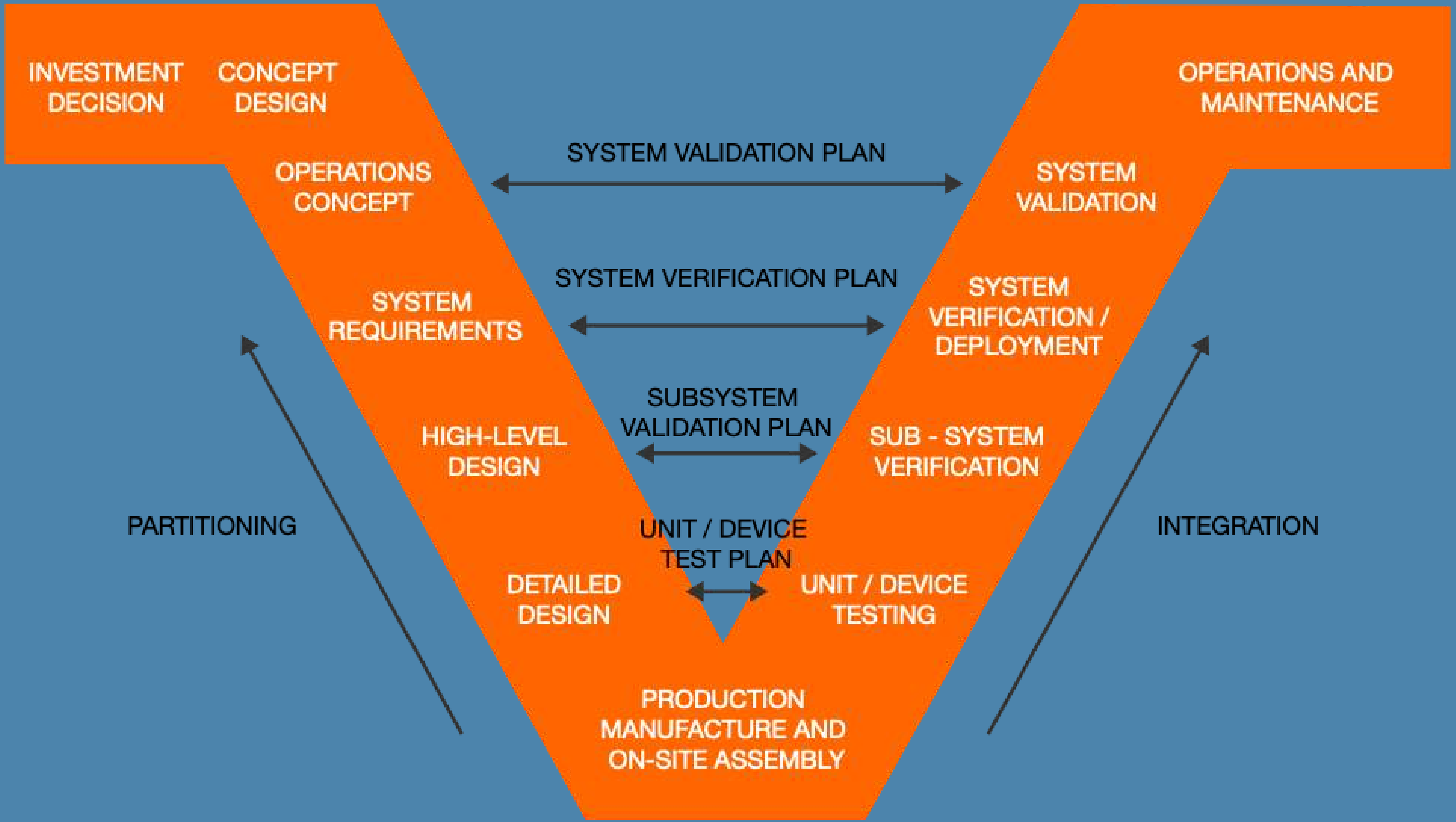
- Process to ensure the drawings and specifications are developed correctly
- “Correct”
  - Fulfill stakeholder requirements
  - Free from errors
  - Constructible
- Tools
  - Regular checks and independent reviews
  - Validation of project requirements against design

DESIGN MANAGEMENT DURING CONSTRUCTION

- Process to ensure we manage changes made to the design during construction
- Manage changes - ensure the design remains "correct"
- Tools
    - RFIs - Construction team requesting changes
    - NCR - Errors lead to changes
    - DCN - Design change notice. Designers directing changes

COMMISSIONING

- Validating that the system functions as intended
  - As opposed to being correctly built
- Systems engineering - designing, integrating and managing complex systems throughout their lifecycles
  - Relates how a system is designed with how it functions


INVESTMENT DECISION
CONCEPT DESIGN
OPERATIONS AND MAINTENANCE
SYSTEM VALIDATION PLAN
OPERATIONS CONCEPT
SYSTEM VALIDATION
SYSTEM VERIFICATION PLAN
SYSTEM REQUIREMENTS
SYSTEM VERIFICATION / DEPLOYMENT
SUBSYSTEM VALIDATION PLAN
HIGH-LEVEL DESIGN
SUB - SYSTEM VERIFICATION
PARTITIONING
UNIT / DEVICE TEST PLAN
INTEGRATION
DETAILED DESIGN
UNIT / DEVICE TESTING
PRODUCTION MANUFACTURE AND ON-SITE ASSEMBLY

COMMISSIONING

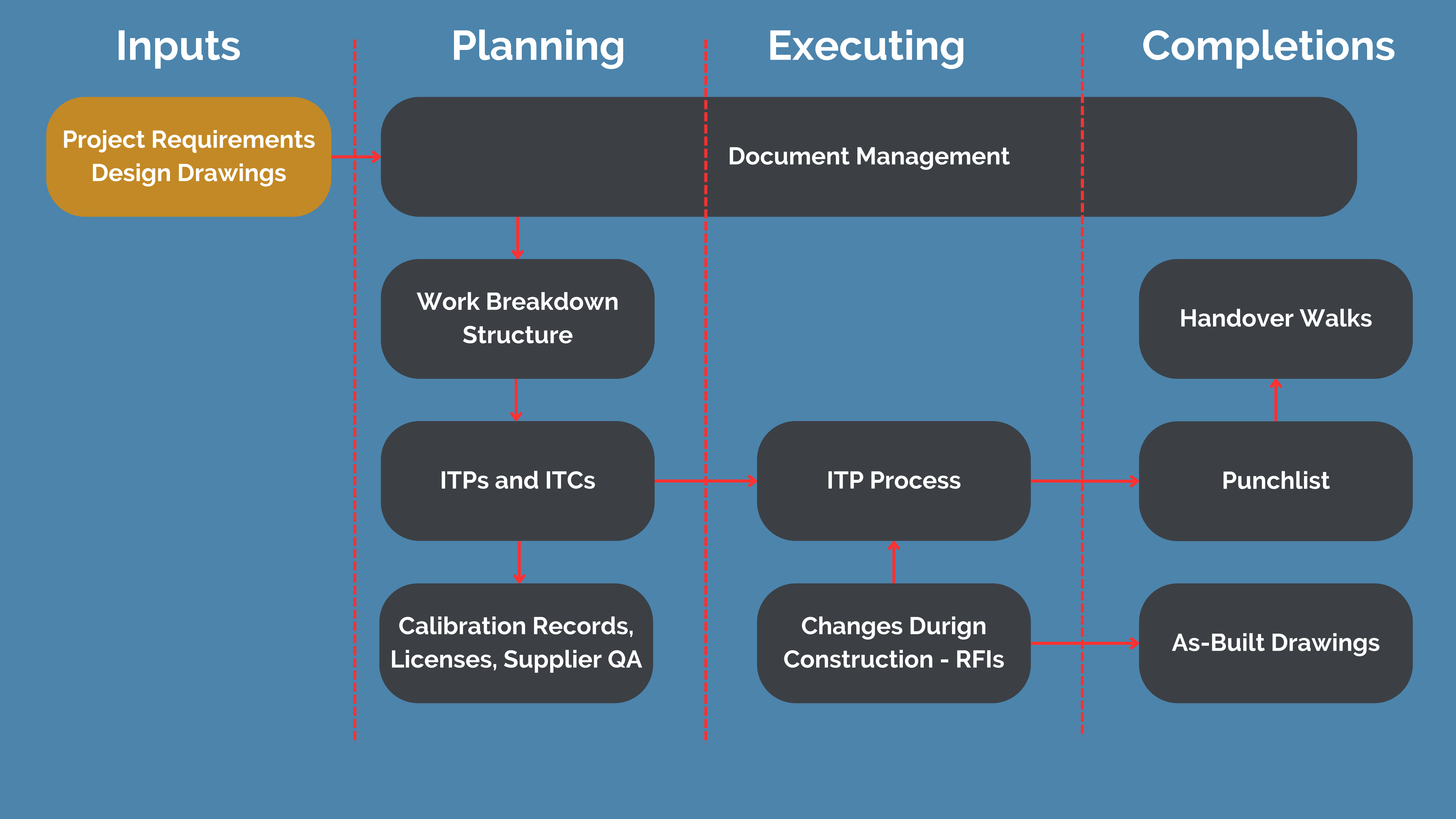
- Construction QA checks that what we have built is correct
- Commissioning checks what we built functions correctly
    - Unit/device level
    - Subsystem validation
    - System verfication

COMMISSIONING - SOLAR FARM

- Project requirements
  - 50MW solar farm
- Design
  - Install 45,000 solar panels
- Construction QA
  - Checks panels are installed correctly
- Commissioning
  - Checks solar farm exports 50MW

CONSTRUCTION QA SYSTEM

- What does a construction QA system consist of?

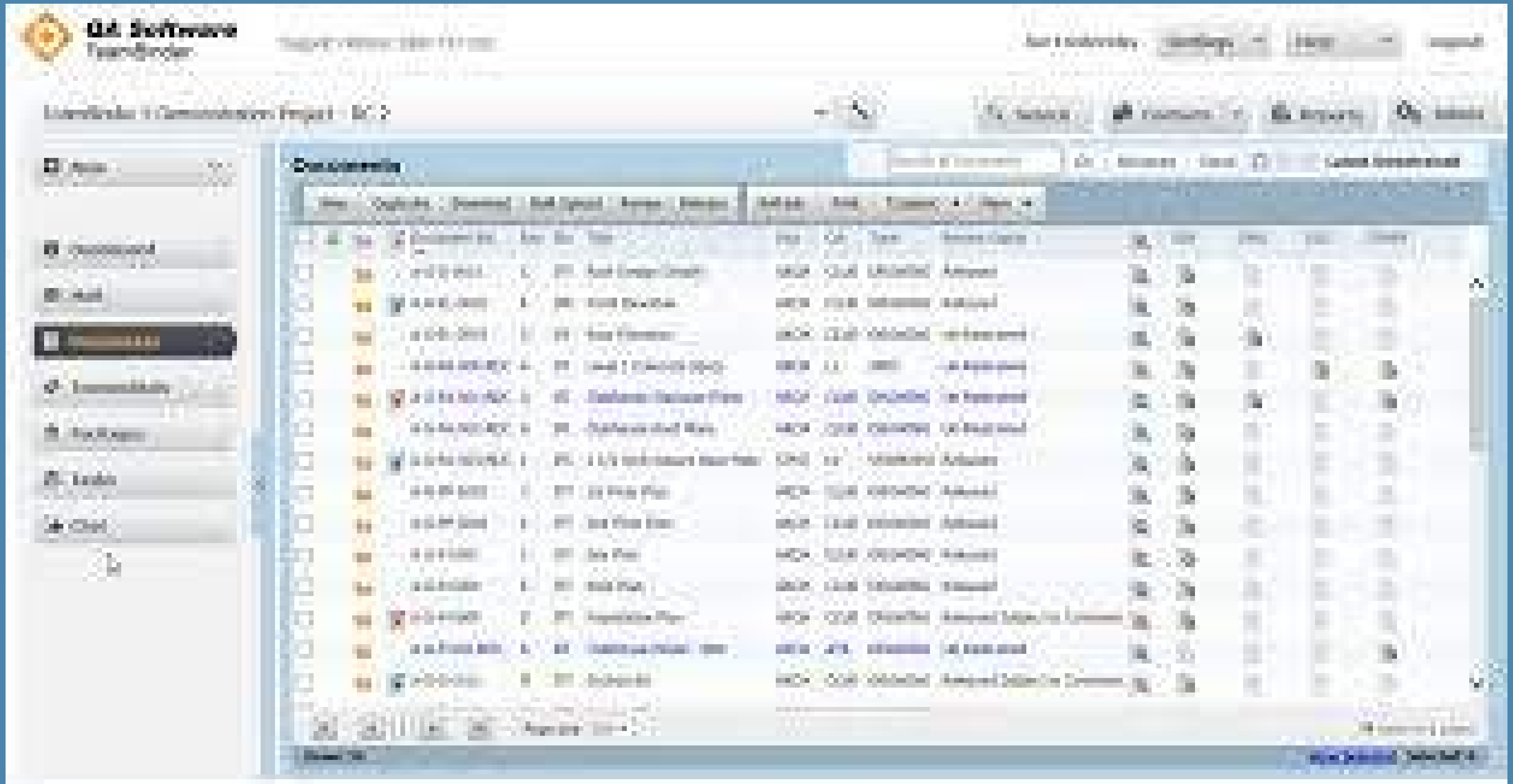

Inputs
Planning
Executing
Completions
Project Requirements
Design Drawings
Document Management
Work Breakdown Structure
Handover Walks
ITPs and ITCs
ITP Process
Punchlist
Calibration Records, Licenses, Supplier QA
Changes Durign Construction - RFIs
As-Built Drawings

PROJECT REQUIREMENTS

- Primary input to process
- Types
  - Design drawings
  - Project specifications
  - Standards and specifications

DOCUMENT MANAGEMENT

- System used to control project documents
- Projects involve 1000s of documents
- Ensure correct revisions, status, etc.



WORK BREAKDOWN STRUCTURE

- Hierarchical decomposition of the project scope
- Break the project scope down into component pieces
- Identify the scope of individual inspection and test plans
    - Ensure the entire project scope is covered
    - No testing activities missed

ITPs - Preparation

- Prepare individual inspection and test plans
- Identify every step in construction process
    - What needs to be checked?
    - What can go wrong
- ITP components
    - Pre-works - materials, licenses, drawings
    - Construction - Inspection and Test Checksheets
    - Close-out - RLMU, NCRs
- Approval from client

Inspection and Test Checksheeets

- Individual checksheets for construction activities
- Step by step documenting the process
  - Every step and every check required to construct as per drawings
- ITC has a specific scope
  - 1 x ITC per day
- Examples
  - Pre-pour checklist
  - Cable testing

Punchlist

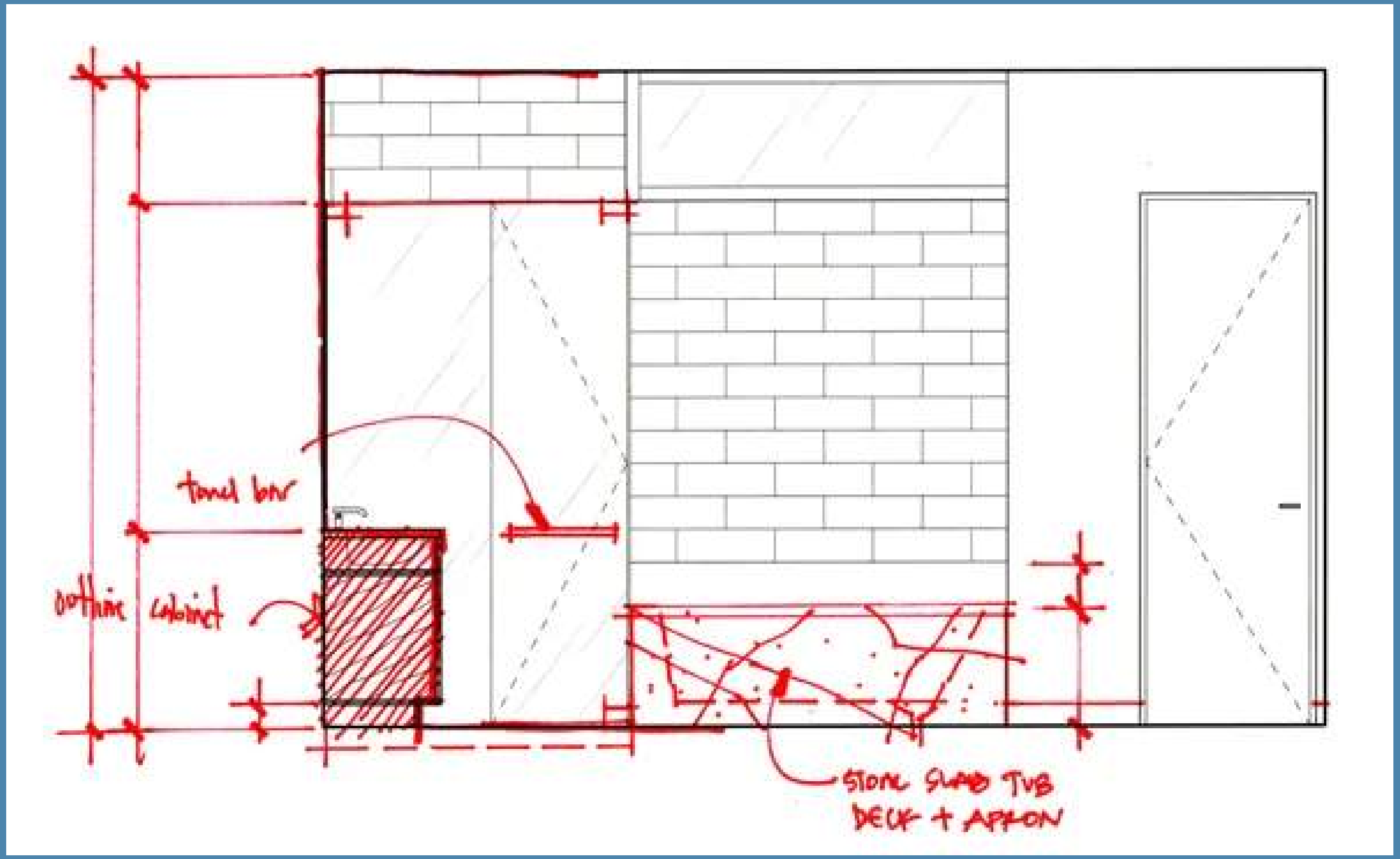
- As ITPs/ITCs are completed, defects will be identified
- Failed inspection and testing - records added to punchlist
    - I.e. cable labels missing from cable
    - Formwork not braced correctly
- Punchlist is a record of all defects/minor outstanding works
    - Categorised appropriately
    - Responsibility allocated

Design Changes - RFIs

- Through construction, there will be changes to design
- Managed through RFIs - Request For Information
  - Formal change management process
  - Approval and direction from design/client
- An important component of QA process
  - Verification that what we are building matches drawings and specifications

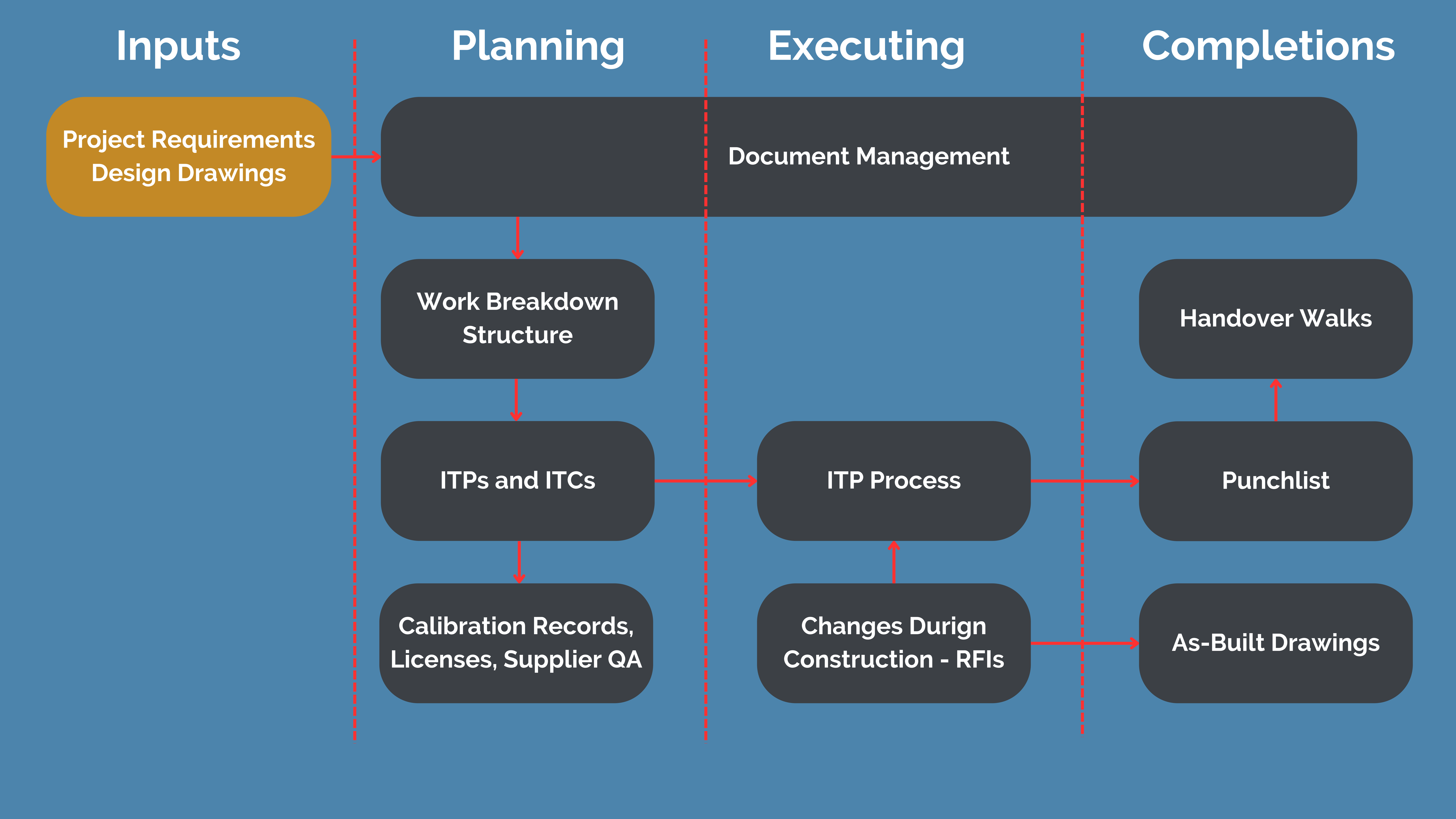
As-Built Drawings

- All changes to design captured on As-Built Drawing set
- Construction team complete Red-Line Mark-Ups
- New drawing revision created



Handover Walks

- Completion of construction works - handover walk-through
- Walk-through with client, owner's engineer
- Inspection of works and opportunity to add to punchlist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Handover Walks
Punchlist
As-Built Drawings

The Biggest Mistake....

- Failing to integrate quality with contract milestones/requirements
- Quality has significant commercial implications
- Impacts payment process - timing and value
- Quality should be integrated with payment milestones and timing
    - Client - opportunity to ensure work is satisfactorily completed
    - Contractor - opportunity to justify value of payment claim

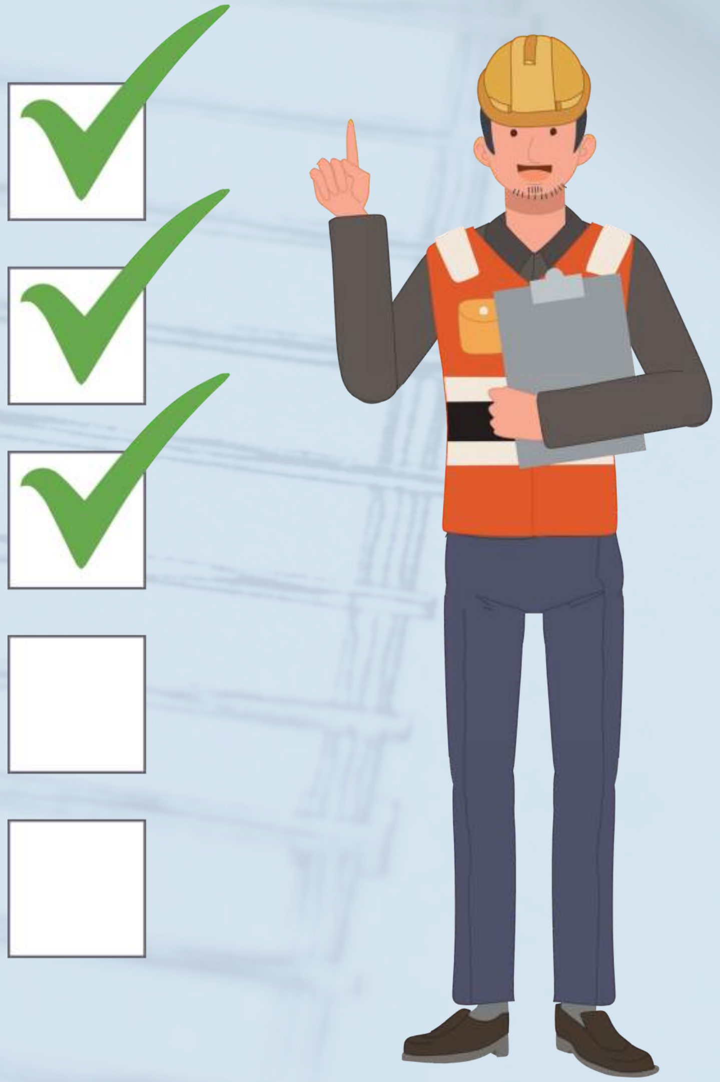
INSPECTION AND TEST PLANS

Outlines the procedures, tests, and inspections required to ensure that a construction project meets specific quality standards and requirements

INSPECTION AND TEST PLAN

Concrete pour: Concrete mix, placement method, formwork stripping time, etc.

INSPECTION AND TEST PLAN



Every step in the process and how we ensure its done correclty

INSPECTION AND TEST PLAN

- All significant construction activities
  - Piling
  - Conduit Installation
  - Concrete Pour
  - Traffic management?
- Prepared before construction works
- Approved by client

ITP STRUCTURE

Inspection and Test Plan

Details

Project Name	Western Road Project	**Date**	
Inspection and Test Plan	Installation of Electrical Conduit	**Engineer Responsible**	
Work Package		**Sub-Contractor (If Applicable)**	
Design Package		**Relevant Standards/Specifications**	AS3000 Western Road Project Services Specification

Symbol	**Description**
H	Mandatory Hold Point
W	Witness Point

ITP STRUCTURE

`Inspection and Test Plan Template

Test Activity	Relevant Specification	Test Method	Timing/Frequency	Acceptance Criteria	Responsible
1. Pre-Works					
2. Construction					
3. Close-Out					

ITP STRUCTURE

List out all steps

`Inspection and Test Plan Template

Test Activity	Relevant Specification	Test Method	Timing/Frequency	Acceptance Criteria	Responsible
1. Pre-Works					
2. Construction					
3. Close-Out					

ITP STRUCTURE

Governing Specification (i.e. AS3000)

`Inspection and Test Plan Template

Test Activity	Relevant Specification	Test Method	Timing/Frequency	Acceptance Criteria	Responsible
1. Pre-Works					
2. Construction					
3. Close-Out					

ITP STRUCTURE

Test we are going to conduct

`Inspection and Test Plan Template

Test Activity	Relevant Specification	Test Method	Timing/Frequency	Acceptance Criteria	Responsible
1. Pre-Works					
2. Construction					
3. Close-Out					

ITP STRUCTURE

How often are we going to do this test?

`Inspection and Test Plan Template

Test Activity	Relevant Specification	Test Method	Timing/Frequency	Acceptance Criteria	Responsible
1. Pre-Works					
2. Construction					
3. Close-Out					

ITP STRUCTURE

Evidence the activity passed the test

`Inspection and Test Plan Template

Test Activity	Relevant Specification	Test Method	Timing/Frequency	Acceptance Criteria	Responsible
1. Pre-Works					
2. Construction					
3. Close-Out					

ITP STRUCTURE

Who is responsible for approving?

`Inspection and Test Plan Template

Test Activity	Relevant Specification	Test Method	Timing/Frequency	Acceptance Criteria	Responsible
1. Pre-Works					
2. Construction					
3. Close-Out					

INSPECTION AND TEST PLAN

- All significant construction activities
    - Piling
    - Conduit Installation
    - Concrete Pour
    - Traffic management?

INSPECTION AND TEST PLAN

- Prepared before construction works
- Approved by client

WRITING

1. Review drawings/specifications
2. Develop construction methodology
3. List out all the steps in the process.
4. Identify
    a. What detail/information does the drawing show?
    b. What can go wrong?
5. Fill out supporting information
6. Submit to client for approval

WHAT CAN GO WRONG?

- Incorrect survey set-out
- Wrong concrete material used
- Wrong placement method
- Not compacted/vibrated properly
- Formwork not properly braced
- Reinforcement moves during pour
- Cast-in items not installed
- Cast-in items not secured

These are the things we need to check!!

CONSTRUCTION METHODOLOGY

- **Pre-works**
  - **Design** - drawings/specifications reviewed and correct. Any RFIs current
  - **Material supply** - approved materials procured, factory testing, dockets, etc.
  - **Handover** - handover from previous trades. QA complete
  - Survey - Survey set-out of works
- **Construction**
  - Steps in the construction process
- **Close-out**
  - **Testing** - final testing
  - **As-built** - survey as-builts. Red-Line Mark-Ups
  - **Handover** - hand-over to other trades

SPECIFICATIONS

- Governing specifications/requirements
- Types
  - Project specifications
  - Drawings
  - Standards

C14. CONCRETE PROPERTIES SHALL BE AS FOLLOWS U.N.O.:

ELEMENT AND LOCATION	GRADE (MPa)	MAX. W/C RATIO	MAX. SLUMP (mm)	MIN. CEMENT CONTENT kg/m^3	MAX. 56 DAY DRYING SHRINKAGE (MICROSTRAIN)
FIBRE REINFORCED FOOTINGS	32	0.5	100+/-20	310	N/A
FOOTINGS	25	0.56	100+/-20	290	N/A
OFFICE SLAB	32	0.5	80+/-15	310	600
WAREHOUSE SLAB	40	0.45	80+/-15	360	600
EXTERNAL HD SLAB	32	0.5	80+/-15	310	600

TEST METHOD

- Details of testing being undertaken
- Specific discipline/activity
  - Concrete testing
  - Compaction testing
- Generic
  - Inspection
  - Photos
  - Dockets

TIMING/FREQUENCY

- How often/when will testing be completed
- Time-based
  - Every day/week
- Activity based
  - Every lot
  - Every 20m of trenching
  - Per concrete pour

ACCEPTANCE CRITERIA

- Require result that test/inspection needs to achieve
  - 32MPa 28 day concrete strength
  - >50MOhm insulation resistance
- Types
  - Test results
  - Subjective verification
  - Records
  - Design criteria

RESPONSIBLE

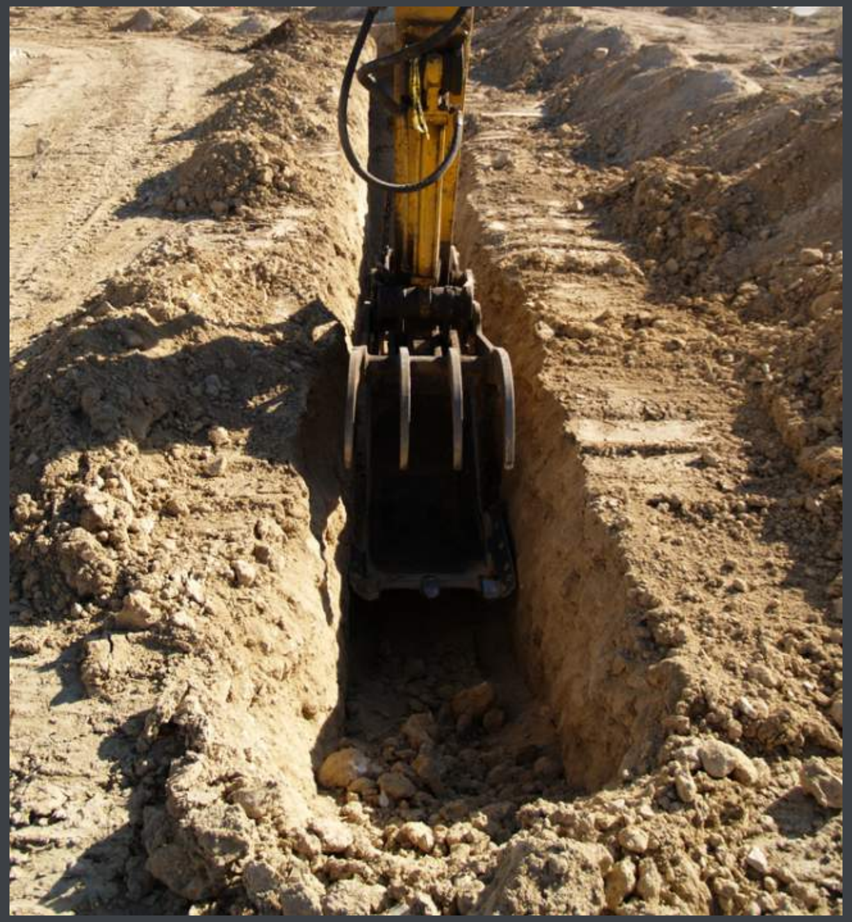
- Who is responsible for approving the works?
- Parties
  - Sub-contractor
  - Engineer
  - Supervisor
  - Client/Owner's Engineer
  - Stakeholder

RESPONSIBLE

- **Hold point** - works cannot proceed past this point
- **Witness point** - advised but can still proceed
- **Review point** - advised after the fact

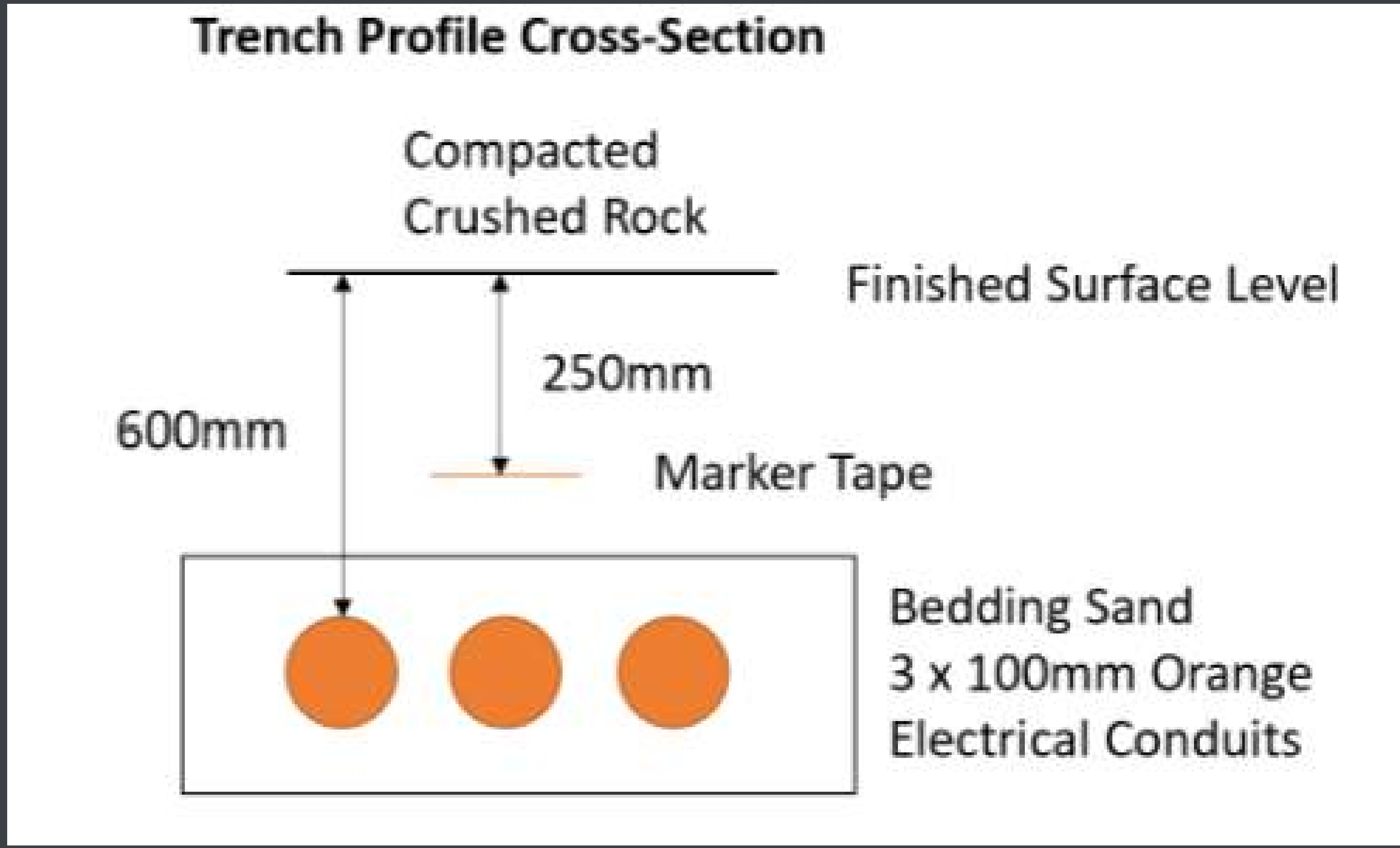
EXAMPLE - CONDUIT INSTALL

- Installation of underground electrical conduits



INPUTS

- **Design**
- **Methodology**
  - Excavate
  - Lay conduit
  - Backfill
- **Specification**
  - Approved materials
  - Excavation permit
  - Compaction testing
- Mandrel testing



STRUCTURE

`Inspection and Test Plan Template

Test Activity	Relevant Specification	Test Method	Timing/Frequency	Acceptance Criteria	Responsible
1. Pre-Works					
2. Construction					
3. Close-Out					

STRUCTURE

Test Activity	Relevant Specification	Test Method	Timing/Frequency	Acceptance Criteria	Responsible
1. Pre-Works					
Design	Issue For Construction Drawing Set	R	Prior to Works Commencing	IFC Drawings Approved RFIs	PE - H
Permits	Excavation Permit	R	Prior to Works Commencing	Approved Excavation Permit	PE - H
Survey	IFC Survey Model Excavation Permit	D	Prior to Works Commencing	Trench alignment and existing services marked out	SE - W
Handover from previous trades	N/A	V	Prior to Works Commencing	Work area clean, no obstructions	SP - RA
2. Procurement of Materials					
Conduits	WRP Services Specification AS3000 IFC Design	R	Each Delivery	Type approval for materials Delivery Dockets	SE - W
Marker/Warning Tape	WRP Services Specification AS3000 IFC Design	R	Each Delivery	Type approval for materials Delivery Dockets	SE - W
Backfill	WRP Services Specification	R	Each Delivery	Type approval for materials Delivery Dockets	SE - W
3. Construction					
Trench Excavation - Excavated to correct depth - Location	WRP Services Specification AS3000 IFC Design	D	Every 20m Prior to conduit installation	Survey confirmation of depth below Finished Surface Level	SE-W
Conduit Installation - Correct size and no. conduits - Conduits glued correctly - Spacing between	WRP Services Specification AS3000 IFC Design	R/D	Every 20m Prior to Back-Fill	Photos showing conduit arrangement as per design Survey Pick-Up of Conduits prior to back-full	SE-W

APPROVALS

- Internally
  - Project/Senior Project Engineer
  - Quality team
- Externally
  - Client
  - Owner's Engineer
- Close out all comments before use

NEXT STEP?

- Complete ITPs in-line with construction activities!
- Part of quality management system

DEFECTS

- Definition
- Why they matter
- Types
- Management

DEFINITION

DEFECTS

Failure to meet specified requirements or standards

EXAMPLES

- Poor line marking on a road project
- Missing labels on electrical cables
- Cracking in concrete structure
- Water ingress into the basement of a building
- Missing torque marks on bolts

DEFECTS

Failures to meet the agree upon project requirements and standards

DEFECTS

Elimination of defects - goal of the Quality Assurance process

WHY DEFECTS MATTER

- The 80/20 rule
- 80% of the work takes 20% of the effort
- The final 20% (more like 5%) takes 80% of the effort

WHY DEFECTS MATTER

- Safety
- Value
- Customer/Stakeholder satisfaction
- Contracts
- Cost
- Time

SAFETY

- Origination of design requirements often safety
  - Concrete strength
  - Insulation of electric cables
- Ensure construction complies with project specifications
  - Ensure its “safe”

SAFETY - SURFSIDE CONDOMINIUM COLLAPSE

- Failed waterproofing on construction of a 12-story building
- Corrosion of reinforcement
- Damage over time lead to the collapse of a structure
- Death of 98 people

VALUE

- Defects are failures to meet project specifications
- Specifications originate from requirements
- Requirements are the components of the project that generate value
- More defects = less value

STAKEHOLDER SATISFACTION

- Defects impact the perception of product
- Aesthetic/minor defects particularly
- Project still “functionally” works, however poor perception

CONTRACTS

- Contractor's ultimate responsibility is to deliver the project in accordance with specifications and requirements
- Requires product to be free from defects
- Contractual implications
  - Payment
  - Contract completion - return of security

CONTRACTS - PAYMENT

- Payment is always an issue on construction projects
- Principal pays contractor based on work they have completed
  - Disputes always arise around incomplete works and valuation
- Defects can be subjective

COST

- Rework costs money
- Cheapest way to do a job
  - Do it once
  - RFT - Right First Time
- Defects
  - Incomplete work
  - Re-work

TIME

- Extension of cost
- Quickest way to complete a project
    - RFT - right first time

CATEGORIES OF DEFECTS

CAUSE

Design, construction, materails, missing equipment, etc.

SIGNIFICANCE

Aesthetic defects or significant to operation of asset

RESPONSIBILITY

Who is responsible for rectifying the defects

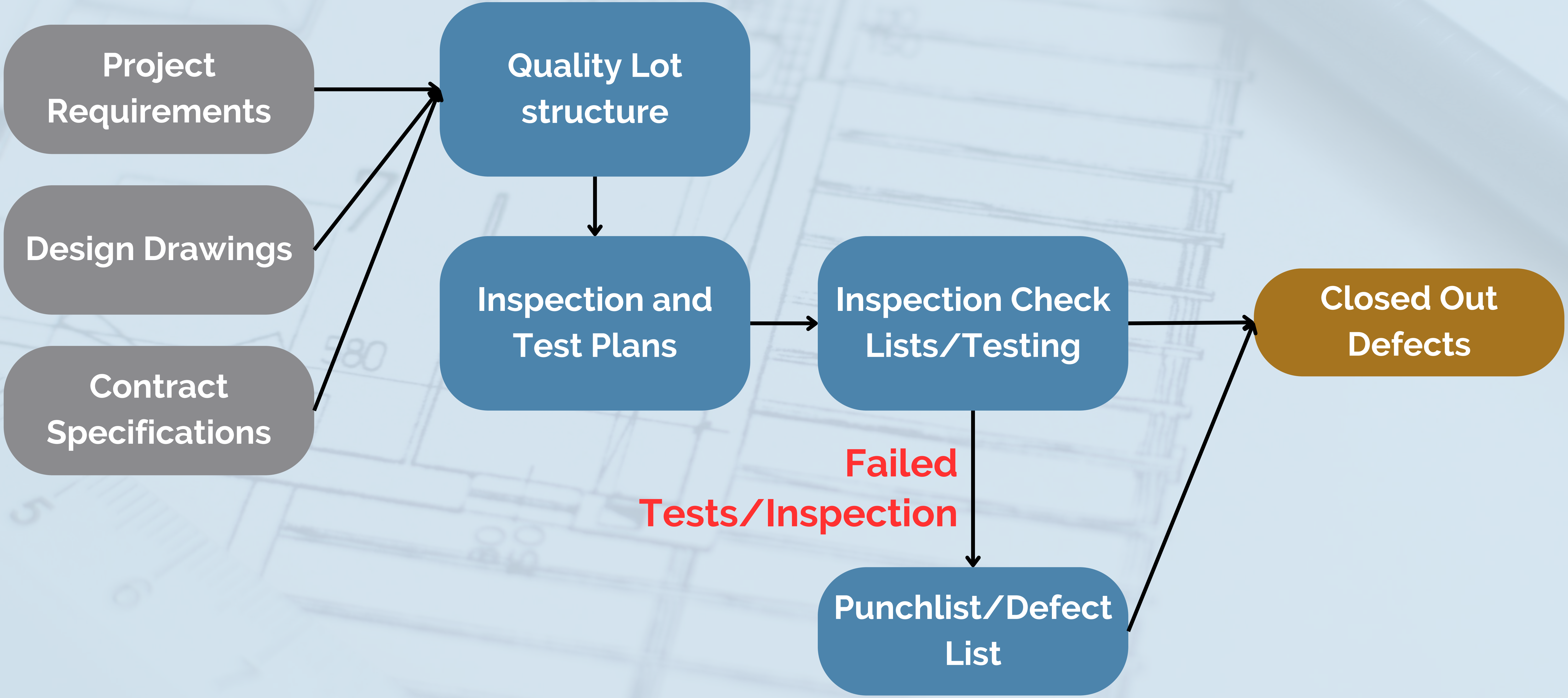
TYPE

Incomplete work vs incorrect installation

HOW DO WE MANAGE DEFECTS?

- Quality management process exists to eliminate defects
  - Deliver a product that fulfils requirements
  - The process identifies and then closes out defects

QUALITY MANAGEMENT PROCESS



NCRs

- What an NCR is
- NCR vs Defect
- Components of an NCR
- Preparing an NCR
- Example

DEFINITION

NON-CONFORMANCE REPORT

Formal document used to identify, track, and resolve instances where a project does not conform to specified standards, requirements, or plans

NON-CONFORMANCE

Defect - Failure to meet requirement

Significant

Non-conformance Report

DEFECT VS NCR

- Defect is any failure to meet project requirements
  - Minor or major
- Minor - missing labelling on electrical cable
- Major - incorrect cable specified in design requiring major re-work
- NCR - Used to capture significant defects
  - Depends on project/company specifics

WHEN DO YOU NEED AN NCR?

- Significant defect
  - Commercially
  - Time
  - Cost
- Investigation required
- Change to original standards

NCR Components

IDENTIFY

Identify and define the NCR

ANALYZE

Work out what the cause was

RESOLVE

Determine what we are going to do now

PREVENT

Avoid any further non-conformances

IDENTIFY

- NCRs identified during:
    - Inspection and Testing
    - Audits
- Can be identified by
    - Client
    - Contractor
    - Sub-contractors

ANALYZE

- Identify the cause of NCR
- Responsibility - commercial liability
  - Who is responsible for the NCR
- Why did the NCR happen in the first place?
  - Design
  - Materials
  - Workmanship

RESOLUTION

01. RE-WORK TO MEET STANDARD

Re-do the activity to meet the original requirement

02. ACCEPT

Accept without any repair

03. REPAIR WITH CONCESSION

Repair to new standard and requirement

RESOLUTION

Client approval required

01. RE-WORK TO MEET STANDARD

Re-do the activity to meet the original requirement

03. REPAIR WITH CONCESSION

Repair to new standard and requirement

02. ACCEPT

Accept without any repair

EXAMPLE NCR

Non-Conformance Report

Non-Conformance Report

Details

Include screen recording

Project Name	Western Road Project	**Date**	27/05/21
NCR Name	Under-Bore Missing Trace Wire	**Engineer Responsible**	Tim Fairley
Work Package	Combined Services Trench	**Sub-Contractor (If Applicable)**	Under Boring Solutions
Design Package and Drawing Reference	WRP-Services Drawing 2178	**Relevant Standards/Specifications**	AS3000 Western Road Project Services Specification
Relevant Quality Lot	CST -> Area 1 -> Jones Road Under Bore		
NCR Description	The standard design details for the under detailed in Drawing 2178 show that each underbore requires 3 x 100mm electrical conduits and 1 x 100mm communications conduits. Additionally, a 5mm steel tracer wire needs to be installed external to the conduit. This tracer wire is used so that, in the future, when service proving, the location of the bore can be mapped using a service locator. On 20/05/21, Under Boring Solutions constructed the bore under Jones Road. They drilled the pilot bore and pulled through 3 x 100mm electrical conduits and 1 x 100mm communications conduit. However, they did not pull through the tracer wire. The bore was sealed with grout and it would now require the construction of a new bore to install the tracer wire. This tracer wire would also need to be off-set from the actual bore location. This inhibits the functionality of the tracer wire in providing service proving benefits.		
Type of NCR	Product NCR		

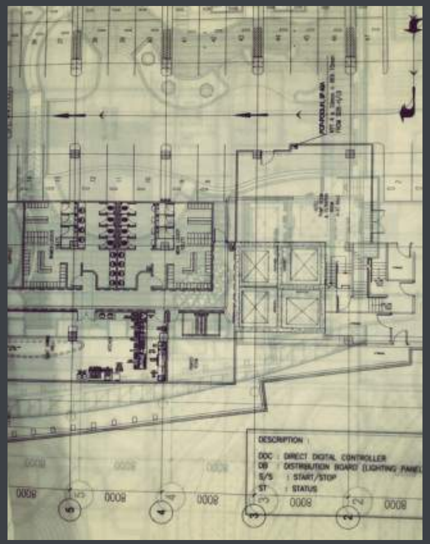
Remedial Action Proposed

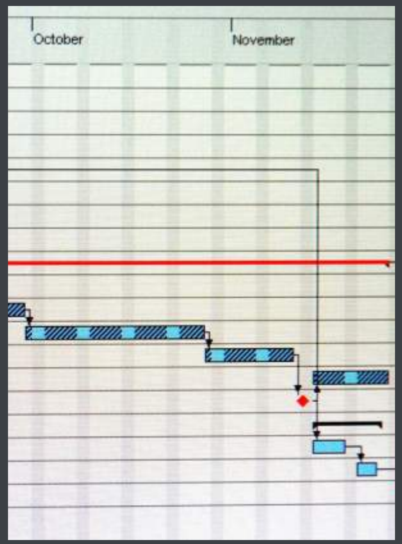
Proposed Rectification	Repair with Concession
Rectification Details	Install 5mm steel tracer wire within one of the spare 100mm electrical conduits. The under bore has been constructed to feed 1 x Distribution board on the east side of Jones Road. Therefore, only 1 of the 100mm conduits is required to run cables in it leaving 2 x spare 100mm electrical conduits in the bore. The construction team propose installing the 5mm steel tracer wire in one of the spare conduits. This tracer wire will be labelled and terminated in the electrical pit.
Primary Cause of Non-Conformance	People

Purpose: What's involved in completing a construction project so you can better plan and prepare

Proof: KPMG found 69% of construction projects fail to be delivered on time, often caused by defects and rework

PROJECT COMPLETIONS





- What is completions?
- Why it matters?
- Quality vs Completions
- Types of completion
- Completion requirements
- Tips to achieve completions

PROJECT COMPLETIONS

Project Completion: How do we hand over the finished product to the client?

PROJECT COMPLETIONS

Client wants to make sure they get what they paid for and there is a smooth transition to the next phase!

DEFINITION

PROJECT COMPLETIONS

Process through which deliverables are finished, inspected and handed over to the client. Ensuring project meets contract requirements, quality standards, etc.

WHY DOES PROJECT COMPLETIONS MATTER?

COST

- Cost
- Poorly managed completions lead to:
    - Increased defects and rework
    - Extended allocation of project resources
    - Liquidated damages and penalties

CONTRACTS

- Contractual requirements
  - Contracts specify completion on a certain date
  - Completion obligations specified
- Contractually obliged to meet these dates

CLIENT

- On-time completion (per requirements) is one of the key objectives the client values
- The client is able to use asset
- Incomplete projects damage contractor’s reputation
    - Secure new work

PROJECT COMPLETION AND QUALITY MANAGEMENT

PROJECT COMPLETION AND QUALITY MANAGEMENT

- **Quality Management** - Verification and validation that what we have built matches the project requirements
- Project completion requires that quality activities have been completed
  - Defects identified and corrected
- Quality must be done to achieve project completion

TYPES OF PROJECT COMPLETION

- Several different types of project completion exist
- Vary depending on
  - Degree of outstanding works
  - Acceptable for use

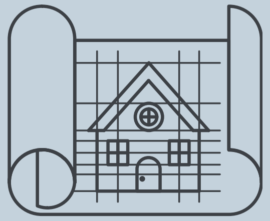
COMPLETION

SUBSTANTIAL

Operational and client can use facility, minor defects

FINAL

The project is fully complete, with no punch list. Final payment



CONDITIONAL

Completion occurs under certain conditions. I.e. temporary occupancy

PARTIAL

Certain parts handed over for use

WHAT'S REQUIRED FOR COMPLETION?

- Sources of requirements
- Types of requirements

COMPLETIONS REQUIREMENTS

CONTRACT MODEL

Responsibility for maintenance, DLP

PROJECT SPECIFICS

Type and nature of project

CLIENT REQUIREMENTS

Specific requirements of the client and systems

INTERNAL

Internal processes and checks

COMPLETIONS

- Construction complete
- Construction QA
- Handover walks
- Commissioning
- Defects
- As-built drawings
- Asset register
- Spare parts
- O&M manuals
- Training
- Maintenance/DLP

CONSTRUCTION COMPLETE

- Physical completion of works
- Functional use of asset
    - Certificate of occupancy
    - Export power on solar farm

CONSTRUCTION QA

- Construction quality assurance complete
- Inspection and testing
- Documented and recorded
- Hold/witness points
- Compile and submit documentation

HAND-OVER WALK-THROUGHS

- Defect inspection with client, architect, etc.
- Opportunity to add to punch list
- Identification of issues
    - Design issues often arise

COMMISSIONING

- Function test installed systems
- Prove system functions as intended
- Additional punch list items
- Witnessed testing

DEFECTS

- Identified during inspection and testing, commissioning, handover walk-throughs, etc.
- Close-out and resolve punch list activities
- Outstanding, minor, cosmetic after substantial completion

AS-BUILT DRAWINGS

- As-built drawing revision
- Red-line mark-ups of IFC design with RFIs, design changes
- Survey as-builts
- New revision of design

ASSET REGISTER

- Documents/records of everything installed
  - Serial numbers
  - Manufacturer
  - Lead times
  - Supplier contacts

SPARE PARTS

- Spare parts to be swapped out during faults
- Specified in contract
  - 5%
  - Vendor specifications
- Special tools/maintenance equipment

O&M Manauls

- Preparation of O&M manual to run the plant
- Detailed maintenance instructions
  - Frequency
  - Types
  - Corrective/prescriptive
  - Fault finding
  - Supplier contacts

Training

- Training session to familiarise O&M team with plant
  - How to use
  - Types of faults
  - Logins/passwords, etc.

Maintenance/DLP

- The contractor may be responsible for maintenance for a set period
- DLP - defects liability period
- Faults arising from construction issues

Other

- Sustainability
- Green star
- Building/plant tuning
- Permits/Certificate of occupancy
- Warranty information

We need to submit everything to complete the job!

TIPS

Treat completions as an individual exercise

Document requirements early

Submit progressively